CHE 494 Final Project

Biopharmaceutical Process Development & Manufacturing

Team 3: **Bailin Feng, Maryam Abdul-Kader**

1. What makes you excited (or not) to work for biopharmaceutical company?

Maryam

Working in biopharma means staying at the forefront of industry innovation while knowing the work I do directly supports human health.

As a biomedical engineer with a focus on cell & tissue engineering, I'm particularly drawn to the focus on biologics, peptides, and mRNA technologies, and the opportunity to contribute hands-on to that work.

Bailin

- Apply my skill sets to make the impact on the market
- In vivo CAR-T/ CAR-Mac
- Gene editing

abbvie Lilly MERCK NOVARTIS novo nordisk® sanofi

2. What is the most common purification technique for the large molecule drug substance?

Chromatography

- Tangential flow filtration(TFF)

Why?

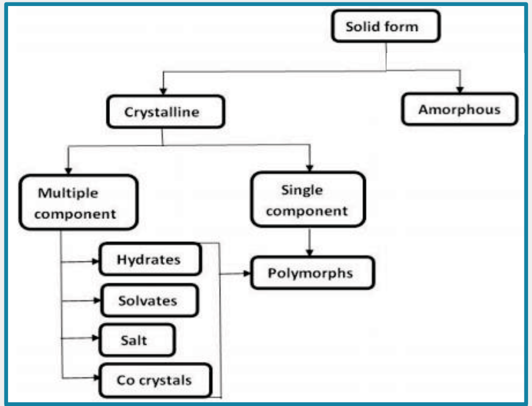
- It separates product by size without stressing fragile molecules
- flow is parallel (tangential) to the membrane rather than perpendicular. This reduces fouling and shear stress compared to dead-end filtration.
- Scalable, automation, rapid purification, versatility in Downstream Processing

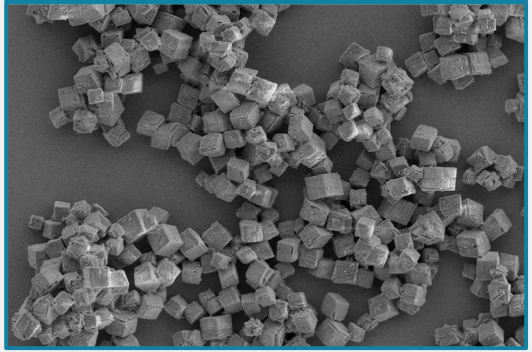
2. Key considerations for process development and reliable scale-up

Key Considerations

- **GMP (Good manufacturing practice)**
- Column Geometry & Scale-Up Rules
- Resin Selection & Lifetime
- Yield vs. Purity Tradeoff
- Cleaning & Sanitization (CIP)
- Process Analytical Technology (PAT)

3. What are the key considerations while selecting the crystal form of the small molecule drug substance? Why?





Key Considerations[4]

- High thermodynamic/chemical stability
- Optimized solubility & bioavailability
- Consistent manufacturability
- Others: Hygroscopicity, Solid Form Screening

Why?

- Prevents polymorph conversion during storage, ensuring consistent potency and dissolution over shelf life
- Different polymorphs of the same drug can have vastly different solubility, directly impacting therapeutic efficacy
    - The most soluble form is often the least stable, an optimal balance must be struck
- The crystal form must survive all unit operations (milling, granulation, compression) without converting, or risk altering critical product quality attributes

4. What is the most challenging small molecule drug product process? What are the challenges and how to make this drug product process work?

Poor Aqueous Solubility

- ~90% of pipeline compounds are poorly soluble
- Low and erratic dissolution limits drug absorption
- Enabling formulations (ASDs, nanosuspensions) are complex and costly
- Solutions often introduce new instability issues

Strategies

- Amorphous Solid Dispersions (ASDs)
- Salt Formation
- Lipid-Based Drug Delivery (LBDDS)
- Nanosuspensions

Other Challenges

- Low bioavailability
- Manufacturing HPAPIs
- Solid-state control & Polymorphism
- Scale-Up & Process Analytical Technology (PAT)

5. What is the simplest small molecule drug product process? What are the challenges to make this drug product process work?

Blend API+ excipient, compress into tablets, package

API
1. poor flow → inconsistent die filling
2. Poor compressibility → weak tablets
3. Poor bioavailability → low efficacy

Tablets
1. Stability over shelf life
2. Scale up

6. What would be your considerations while designing and scaling-up of mixing processes for biologics molecules? Why?

1) Shear sensitivity vs. mixing efficiency

high shear stress damage the biologics molecule structure integrity

2) Air–liquid, liquid-solid interface and foaming

Too much vortexing could cause aggregation at interface

3) Temperature control

biologicals are temperature sensitive

4) Control Reynolds number

Reynolds number determine mixing time, concentration determine nucleation time

7. What drug product process would you recommend fo solid dosage form?

A poorly soluble small molecule drug substance (active pharmaceutical ingredient, API) has very bad physical properties, namely the broader particle size distribution, poor flowability and low bulk density. What drug product process would you recommend for solid dosage form? Provide a process flow chart and outline process design considerations for the unit operations involved.

Dispensing → Premixing → polymer/lipid Solution → Flash nanoprecipitation

Drying → Milling & Sizing → Final Blending → Tablet Compression → Packaging

8. What is a common drying process used for the large molecule drug product?

lyophilization

- Cryoprotectant
- Glass Transition Temperature, Tg ;Melting Point,Tm
- Rapid freezing at - 80 °C

Mass transfer, heat transfer and freezing step for scale up